Worksheet: Database Normalisation to Third Normal Form (3NF)

Scenario: Riverside Community Market

Purpose of This Activity

In this worksheet, you will analyse a **poorly designed database table** and explain how it can be **normalised to Third Normal Form (3NF)**.

By the end of this activity, you should be able to:

- Identify **problems** in an unnormalised data set
- Explain **why** normalisation is required
- Describe the steps needed to reach **1NF, 2NF, and 3NF**
- Produce a **clear written justification**, suitable for exam answers

Important: Marks are awarded for **explanation and reasoning**, not just drawing tables.

The Original Data (UNF)

Context

The table below is used by Riverside Community Market to store **customer orders**.
It was created quickly and **not designed using normalisation principles**.

Table: RCM_Orders_UNF

Order ID	Customer First Name	Customer Surname	Order Date	Producer Manager ID	Order Status	Products Ordered	Collection / Delivery Location	Customer ID	Manager First Name	Customer Address	Customer Postcode	Total Cost
1	Alex	Brown	01/03/26	12	Preparing	Honey Jars (2), Apple Juice (1)	Market Hub	C01	Sam	12 River Street	QJ2 7YN	£13.50
2	Jamie	Patel	02/03/26	12	Delivered	Cheese Blocks (1), Milk Bottles (4)	Home Delivery	C02	Sam	8 Market Road	MW6 6KZ	£15.75
3	Alex	Brown	03/03/26	15	Confirmed	Honey Jars (1)	Market Hub	C01	Priya	12 River Street	QJ2 7YN	£5.00
4	Chloe	Fields	05/03/26	12	Preparing	Bread Loaves (3), Jam Jars (2), Eggs (12)	Collection Point	C03	Sam	314 Clifton Street	BV5 1OA	£18.20
5	Daniel	Morris	06/03/26	18	Confirmed	Apple Juice (5), Cheese Blocks (2)	Home Delivery	C04	Sam	94 Deans Avenue	FQ4 2PH	£27.00

Step 1: Identify Why This Table Is NOT Normalised

Before normalising, you must **identify the problems**.

Look out for the following:

- Columns that contain **more than one value**
- Data that is **repeated across multiple rows**
- Data that **does not describe the same thing**
- Data that would be difficult to **update accurately**

Discussion Questions

Answer in full sentences:

1. Which column clearly breaks **First Normal Form (1NF)**?
2. Why is storing multiple products in one field a problem?
3. Which data is repeated for the same customer?
4. Which data relates to **customers**, **orders**, **products**, or **managers**?

Step 2: First Normal Form (1NF)

To reach 1NF, a table must:

- Have **no repeating groups**
- Have **atomic (single) values** in every field
- Be able to identify each record uniquely

What to Look For in This Data

- The **Products Ordered** column contains lists and quantities
- One order can contain **multiple products**
- A single field contains **both product names and quantities**

Write-Up Guidance (Use This Language)

When writing your answer, you should explain that:

- The table is not in 1NF because it contains **multi-valued attributes**
- The products need to be stored in a **separate table**
- Each product ordered should appear in its **own row**

Step 3: Second Normal Form (2NF)

`To reach 2NF:`

- The table must already be in **1NF**
- All non-key attributes must depend on the **whole primary key**
- Partial dependencies must be removed

`What to Look For in This Data`

Assume an order with multiple products would require a **composite key**.

Ask yourself:

- Does `Customer First Name` depend on the **order**, or the **customer**?
- Does `Customer Address` depend on the **product**?
- Does `Manager First Name` depend on the **order item**?

`Write-Up Guidance (Use This Language)`

In a strong exam answer, you should explain that:

- Customer details depend only on **Customer ID**
- Manager details depend only on **Manager ID**
- These attributes should be moved to **separate tables**
- This removes **partial dependencies**

Step 4: Third Normal Form (3NF)

To reach 3NF:

- The table must already be in **2NF**
- There must be **no transitive dependencies**
- Non-key attributes must depend **only on the primary key**

What to Look For in This Data

Ask yourself:

- If you know the **Customer ID**, do you also know the address?
- If you know the **Manager ID**, do you need the manager's name in the same table?
- Does one non-key attribute determine another?

Write-Up Guidance (Use This Language)

A high-quality explanation will state that:

- Customer address and postcode depend on **Customer ID**, not Order ID
- Manager name depends on **Manager ID**
- These transitive dependencies must be removed
- Each table should describe **one entity only**

Final Reflection: Writing a Perfect Exam Answer

To achieve full marks, your write-up should:

- Refer directly to **examples from the data**

- Explain **why** each normalisation step is needed
- Use correct terms such as:
    - *repeating groups*
    - *partial dependency*
    - *transitive dependency*
- Clearly justify how the final structure improves:
    - Data integrity
    - Accuracy
    - Maintenance

Exam Tip:

Explaining *why* a table breaks 3NF is often worth more marks than drawing the final tables.